FY4QPE-MSA: An All-Day Near-Real-Time Quantitative Precipitation Estimation Framework Based on Multispectral Analysis From AGRI Onboard Chinese FY-4 Series Satellites

Ziqiang Ma, Siyu Zhu, and Jun Yang

***Abstract*— Accurate and near-real-time rain information at fine scales is critical for forecasting local weather and floods. Traditional classical infrared (IR) cloud-top brightness temperature data alone do not contain sufficient precipitation-related information, and the introduction of visible (VIS) observations limits their applications to daytime. Methods for the efficient and comprehensive utilization of multichannel IR observations for accurately retrieving all-day near-real-time rain rates with consistent high quality warrant further exploration. In this study, we propose an all-day near-real-time quantitative precipitation estimation framework based on multispectral analysis (MSA) from the Advanced Geosynchronous Radiation Imager (AGRI) onboard Chinese FY-4 series satellites; the proposed framework is called FY4QPE-MSA. Multiple IR bands are comprehensively and efficiently considered by adopting the principal component analysis technique to reduce the dimensionality to a few independent features while preserving most of the variations. The main conclusions include, but are not limited to, the following aspects: 1) the MSA from IR channels provides valuable information that facilitates the more accurate delineations of the precipitation occurrences; 2) FY4AQPE-MSA outperforms FY4AQPE-Offical [∼20% gain in the Pearson correlation coefficient (CC), ∼25% gain in the root mean square error (RMSE), and ∼15% gain in the critical success index (CSI)], FY4AQPE-Single (∼10% gain in CC, ∼15% gain in RMSE, and ∼15% gain in CSI), and Precipitation Estimation from Remotely Sensed Information using artificial neural network-cloud classification system (PERSIANN-CCS) (∼30% gain in CC, ∼20% gain in RMSE, and ∼25% gain in CSI); and 3) compared with the international IR-based baseline precipitation product, i.e., PERSIANN-CCS, FY4AQPE-MSA demonstrates no spatial gaps over the Tibetan Plateau. In addition, the results of this study suggest that the proposed general framework is promising and applicable for Chinese FY-4 series satellites in generating all-day near-real-time rain-rate information.**

***Index Terms*— Artificial neural network (ANN), Chinese FY4A, dual-frequency precipitation radar (DPR), infrared (IR), principal component analysis (PCA), probability distribution function (PDF), rain rate, retrieval, satellite.**

I. Introduction

ACCURATE and near-real-time rain information at fine scales is critical to a wide range of applications, including accurate forecasting of local weather and floods to protect human life and property [1]–[5]. Ground-based rain gauges/radars and satellite-based infrared (IR)/passive microwave (PMW)/radar observations are major sources for retrieving surface rain rates [6]. Collectively, because of the sparse networks of ground observations and their limited spatial representatives, especially over remote regions and oceans [7]–[9], quantitative precipitation estimations (QPEs) from space are the only potential promising alternative to ground-based precipitation measurements. QPEs from space could provide large-spatial-scale and even global precipitation estimates with increasingly fine spatiotemporal resolution and quality [10]. Satellite QPEs can be retrieved through observations from various sensors [6], [7], and the most commonly used observations are IR data from geosynchronous Earth orbiting (GEO) satellites [11] and PMW data from low Earth orbit (LEO) satellites [12], [13]. Though PMW data inherently record microphysical information on hydrometeors within the clouds, PMW-based rainfall retrievals do not provide direct estimates with finer resolutions in near-real-time because of their limited spatiotemporal observing frequencies [12], [14], [15]. However, IR-based rainfall retrievals can be generated, in all-day near-real-time, with fine spatiotemporal resolutions (∼2 km, 1 min, or better) [16], which greatly benefits both the monitoring of complete evolutions of local precipitation events with lifecycles shorter than 1 h and the forecasting of flash floods [6], [14], [17].

Together with the new generation of geostationary weather satellites planned by the international satellite community,

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the Chinese Fengyun (FY, meaning "wind and cloud" in Chinese) meteorological satellite series is becoming an important component [18]. FY-4A (launched on December 11, 2016, located at 99.5° E) marks the start of the Chinese new-generation GEO meteorological satellite series FY-4 (the even number denotes the geostationary satellite series), which is also an experimental satellite that carries four payload instruments: the Advanced Geosynchronous Radiation Imager (AGRI; the primary payload), the Geostationary Interferometric Infrared Sounder (GIIRS), the Lightning Mapping Imager (LMI), and the Space Environment Package (SEP) [19]. The FY-4A/AGRI has 14 spectral bands, similar to other international advanced imagers, such as the GOES-16 Advanced Baseline Imager (ABI) [20] and the Himawari-8 Advanced Himawari Imager (AHI) [21]. Similarly, the FY-4A/AGRI works in three flexible scanning modes: full disk (views of the whole Earth as observed from a geostationary satellite), 15 min; China area (3°–55° N, 70°–140° E), 5 min; and target area (1000 km × 1000 km), 1 min. Most excitingly, FY-4B, the first operational satellite of the FY-4 series, has recently been successfully launched (June 3, 2021, located at 123.5° E), with the number of FY-4B/AGRI bands increased to 15 (http://fy4.nsmc.org.cn/nsmc/en/theme/FY4B.html). Progressively, huge benefits are expected for severe weather monitoring, warning, and forecasting in the global Earth Observing System, especially over the Asia–Pacific region, with advanced imaging instruments onboard subsequent FY-4 series satellites, which are expected to provide measurements with higher spatial–temporal–spectral resolutions. In this scenario, an efficient, effective, and operational framework for retrieving all-day near-real-time rain rates for the FY-4 series satellites would be highly valuable and is urgently needed [4], [5], [19], [22]–[24].

Recently, deep learning techniques have become increasingly popular in estimating and predicting meteorological parameters. For instance, Shen *et al.* [25] estimated ground-level PM2.5 directly from satellite top-of-atmosphere reflectance using a deep learning architecture (Ref-PM modeling) based on deep belief networks (DBNs) without considering the physical processes. Shen *et al.* [26] also used improved five-layer-structured DBNs to map air temperature by comprehensively considering space-remote-sensing and ground-station observations, demonstrating the strong potential of this approach in predicting other meteorological parameters. Chen *et al.* [27] developed an improved convolutional long short-term memory (CLSTM) technique by integrating a raining-oriented loss function and the group normalization technique for nowcasting precipitation with radar echo observations. Algorithms for retrieving IR-based rain rates have been similarly improved, from the random forest (RF) to the artificial neural network (ANN) and conventional neural network (CNN) algorithms [1], [16], [28]–[30]. For example, Kühnlein *et al.* [29] investigated the suitability of using the RF algorithm for estimating rainfall rates based on satellite multispectral visible (VIS)/IR data, and Sehad *et al.* [31] explored the SVM to build different models predicting the rain rates in daytime or nighttime. Overall, ANN is the classical and most representative algorithm for retrieving near-real-time rain rates, with a series of precipitation products, e.g., the Precipitation Estimation from Remotely Sensed Information using ANN (PERSIANN) [28] and its upgraded version, the PERSIANN-Cloud Classification System (PERSIANN-CCS) [16], which also has been integrated into the components for generating the Integrated Multi-satellitE Retrievals for Global Precipitation Measurement (IMERG) product. Various explorations to obtain rainfall estimates with finer qualities using more advanced deep learning techniques based on the core algorithm, PERSIANN (e.g., PERSIANN-DNN [32], PERSIANN-CNN [1], [30], and PERSIANN-cGAN [33]), have also been reported. Although various methods have been explored for retrieving rain rates with high spatiotemporal resolutions from geostationary satellites, room exists for the development of a more efficient and effective framework for accurately retrieving rain rates with enhanced spatial–temporal–spectral resolutions.

Under the assumption that colder cloud tops are correlated with higher rain rates, the band at ∼10.7 μm from GEO satellites, representing cloud-top temperature, have been most frequently used to retrieve near-real-time rain-rate information [16], [28], [34]. Given that IR cloud-top brightness temperature data alone do not contain sufficient precipitation-related information [14], one possible direction is to explore the benefits of multispectral channels from GEO satellites for rain detection and estimation. A common choice is to use VIS cloud albedo data. For instance, Behrangi *et al.* [14] developed a multidimensional estimation method (PERSIANN-MSA) to estimate rain rates using multispectral data including VIS data, which limited its applications to daytime. Meanwhile, a few investigators have explored other IR channels. For example, because of the correlations between water vapor (WV) (IR band at ∼6.5 μm) and precipitation, Tao *et al.* [32] designed a deep learning framework for generating accurate precipitation estimates based on bispectral satellite information (WV and cloud-top temperature). Similarly, Wang *et al.* [30] used three precipitation-related IR bands from GOES-13 [i.e., Band 3 (∼6.5 μm), Band 4 (∼10.7 μm), and Band 6 (∼13.3 μm)] to drive a CNN-based framework for retrieving rain rates. The question of how to efficiently and comprehensively utilize the multichannel IR observations is one of the inevitable trends for accurately retrieving near-real-time rain rates with consistent high quality for all-day operational purposes and warrants further exploration, especially in the era of GEO satellites with greatly enhanced spatial–temporal–spectral resolution measurement ability (e.g., the FY-4 series and subsequent satellites).

In this study, we propose an all-day near-real-time QPE framework based on the multispectral analysis (MSA) from AGRI onboard Chinese FY-4 series satellites; this framework is known as FY4QPE-MSA. The multiple IR bands are comprehensively and efficiently considered by adopting the principal component analysis (PCA) technique to reduce the dimensionality to a few independent features while preserving most of the variations. The results obtained using FY4QPE-MSA are also compared with two single-IR-band (∼10.7 μm)-based precipitation estimates—one widely used as the baseline for IR precipitation estimation algorithm

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TABLE I
DETAILED PARAMETERS OF THE EIGHT IR CHANNELS FROM FY-4A/AGRI

Channel Index	Middle Wave Length	Wave range	Spatial Resolution	Temporal Resolution	Catalog	Main applications
7	3.75μm	3.5 ~ 4.0μm (high)	2km	10-15 min	Near-infrared	Clouds, moisture
8	3.75μm	3.5 ~ 4.0μm (low)	4km	10-15 min	Near-infrared	Land surface
9	6.25μm	5.8 ~ 6.7μm	4km	10-15 min	Middle-infrared	Upper-level water vapor
10	7.1μm	6.9 ~ 7.3μm	4km	10-15 min	Middle-infrared	Middle-level water vapor
11	8.5μm	8.0 ~ 9.0μm	4km	10-15 min	Long-wave infrared	Cloud-top phase
12	10.7μm	10.3 ~ 11.3μm	4km	10-15 min	Long-wave infrared	Surface/cloud-top temperature
13	12.0μm	11.5 ~ 12.5μm	4km	10-15 min	Long-wave infrared	Low-level water vapor
14	13.5μm	13.2 ~ 13.8μm	4km	10-15 min	Long-wave infrared	Clouds, air temperature

development, PERSIANN-CCS, and one generated using FY4QPE-MSA (referred to as FY4AQPE-Single)—to assess the quality of the multispectral data. In addition, the current official rain-rate product of FY-4A (FY4AQPE-Official) is also compared with FY4AQPE-MSA to illustrate the performance of the retrieving method proposed in the present study.

II. STUDY AREA AND MATERIALS

A. Study Area

The study area is located at the Northern Hemisphere observation range of AGRI onboard FY-4A (62.0°–135.0° E, 5°–53° N), which covers Southeast Asia and the southwestern region of the Pacific Ocean. We selected this area by considering three main aspects: 1) the Northern Hemisphere observation from FY-4A/AGRI is one of the most important observations for researchers to investigate the weather system over the East Asian Monsoons; 2) compared with the temporal resolution of the full disk observation of FY-4A/AGRI, that of the Northern Hemisphere observation is higher (∼15 min), providing more direct and instant data for monitoring clouds and rain systems; and 3) southeastern China has frequently suffered from extreme rainfall events due to destructive summer typhoons from the western Pacific Ocean, which threatens people's lives and property. Therefore, accurate and near-real-time precipitation information over this region is urgently needed to alleviate the effects of hydrological-related disasters.

B. Multispectral IR Data From Chinese GEO FY4A/AGRI

With the objective of developing an effective and efficient framework for retrieving all-day near-real-time rain rates for FY-4 series satellites, the eight IR bands from FY-4A/AGRI observations are used in the present study, which has already been translated into physical bright temperatures by the National Satellite Meteorological Centre (NSMC); details of these bands are listed in Table I. FY-4A/AGRI has two middle IR bands centered at 3.75 μm with different dynamic ranges: "3.72-μm high" and "3.75-μm low," which cover 260–450 K and 200–340 K, respectively. In general, 3.72 μm is useful for cloud properties/screening, moisture determination, and snow detection; 6.27 and 7.10 μm are sensitive for middle and upper tropospheric WV detection and tracking; the combinations of bands centered at 6.27, 7.10, and 8.5 μm are helpful for retrieving cloud-top height and cloud microphysical parameters (e.g., cloud phase); 10.8 μm is sensitive for determining of cloud/surface-top temperature, which has been mostly used for retrieving IR precipitation estimates; 12.0 μm is useful for estimating low-level WV and moisture; and 13.6 μm is usually used for monitoring clouds and air temperature [18], [30], [35]. All datasets of FY-4A, both real-time and historical, are available to the global community on the NSMC satellite data server website (http://satellite.nsmc.org.cn).

C. GPM DPR Rain Rate Data

Dual-frequency precipitation radar (DPR) is an active radar designed for observing precipitation and its vertical structure; it is installed onboard the global precipitation measurement (GPM) core observatory satellite, which has been launched and in operation since February 2014 [36]. The DPR consists of a Ku-band precipitation radar (KuPR) and a Ka-band precipitation radar (KaPR). The observations based on these systems are mainly implemented by three swaths: normal scans (NSs), matched scans (MSs), and high-sensitivity scans (HSs). In the present study, the near-surface rain rates obtained by the NSs from DPR are used as the true values when training the rain-rate retrieving models because the NSs contain the information from both KaPR and KuPR, which is freely available through the website https://disc.gsfc.nasa.gov/datacollection/GPM_2ADPR_V06.html.

D. PERSIANN-CCS

The most popular near-real-time precipitation product with the finest spatiotemporal resolutions, PERSIANN-CCS (0.04°, 1 h), is used as the international baseline for evaluating the quality of the estimates retrieved in this study [16]. The PERSIANN-CCS is produced using an ANN model and a self-organizing feature map (SOFM) algorithm with the IR band (∼10.7 μm) as the most important driving data. The PERSIANN-CCS can be used to classify clouds into different categories and then retrieve the rain rates via the quantified relationships between brightness temperature and the ground-radar rain rates. The PERSIANN-CCS can be freely obtained through the Center for Hydrometeorology and Remote Sensing (CHRS) data portal (http://chrsdata.eng.uci.edu/).

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E. FY4AQPE-Official

The official precipitation dataset from FY-4A (FY4AQPE-Official) is provided by the National Satellite Meteorological Centre (NSMC), which has the same spatiotemporal resolutions (0.04°, ∼15 min) as the Northern Hemisphere observations of AGRI onboard FY-4A. When generating the FY4AQPE-Official product, the relationships between the brightness temperature from FY-4A and the PMW rain rates from international satellite precipitation products [12] are established by a probability distribution function (PDF) matching algorithm. The FY4AQPE-Official is freely available through the official website of NSMC (http://satellite.nsmc.org.cn/).

F. Ground Observations

The ground observations are provided by the National Meteorological Information Center (NMIC) from datasets that include the precipitation records from ∼60 000 rain gauges, at hourly scale in spring, summer, and autumn, from 2019 to 2020, over mainland China. This rain-gauge precipitation dataset has undergone three levels of quality control: an extreme values' check, an internal consistency check, and a spatial consistency check [2]. The minimum records of rain rates from the ground observations are 0.1 mm/h; thus, the threshold is also set as 0.1 mm/h to discriminate the rain and no-rain events. The ground observation data are available through the official website of NMIC (http://data.cma.cn).

III. Methodology

A. Framework for Retrieving Rain Rates Based on FY-4A Multispectral IR Observations

There are mainly four procedures in the framework for retrieving the rain rates based on multichannel IR observations from FY-4A/AGRI: data preparation, input feature extraction, training the ANN model for retrieving rain rates, and PDF calibration. The corresponding flowchart is shown in Fig. 1. Because the 12th IR channel (∼10.7 μm) is the most classical band for pixel- or patch-based rain-rate retrievals [16], [28], [34], the eight IR channels were divided into two groups: one group that includes the 12th IR channel and another group that consisted of the other seven IR channels. First, the dimensionality of the other seven IR channels was reduced using PCA to obtain the first two principal components [14]. Both the first two principal components (PC1 and PC2) and the 12th IR channel were subsequently used for extracting the moving-window-based local features (see Table II). Meanwhile, the 12th IR channel was used for deciding the raining patches and extracting the patch-based features (see Table II) using the incremental temperature threshold (ITT) algorithm [16]. The samples for training the ANN model were later constructed by spatiotemporally matching the independent variables (including the moving-window-based and patch-based features; see Table II) and the corresponding dependent variable (GPM DPR rain rates), and the parameters in the ANN model were determined. Finally, the rain-rate estimates using FY4QPE-MSA were obtained by adjusting the retrieved rain rates from the ANN model using a PDF calibration algorithm integrating the elevation effects.

TABLE II

Feature Information Used in the Proposed Framework for Retrieving the Rain Rates on the Basis of IR Observations From FY-4A/AGRI

Features extracted using the moving window strategy			
Window Size	Ch 12	PC1	PC2
1×1	T	PC	PC
9×9	Mean(T)	Mean(PC)	Mean(PC)
15×15	Mean(T)	Mean(PC)	Mean(PC)
Features extracted using the ITT algorithm			
	220K	235K	253K
Patch	Area(Cloud)	Area(Cloud)	Area(Cloud)
Patch	Min(T)	Min(T)	Min(T)
Patch	Mean(T)	Mean(T)	Mean(T)
Patch	Std(T)	Std(T)	Std(T)
Pixel	Distance(Pixel)	Distance(Pixel)	Distance(Pixel)
Pixel	Delta(T)	Delta(T)	Delta(T)
Pixel	Slope(Pixel)	Slope(Pixel)	Slope(Pixel)

B. Description of the Core Algorithms in This Framework

1) Principal Component Analysis: PCA [14] can be used to compress the multidimensional features into variables with a few dimensions; thus, it can delete redundant information and accelerate the model training and retrieving processes. In this study, coefficients of PCA, trained at the seasonal scale, are implemented on seven IR channels except for the 12th channel (∼10.7 μm) to compress the information into the first two main principal components, which preserve 97% or more of the total variations.

2) Input Feature Extraction and Incremental Temperature Threshold Algorithm: Given that the local textural features around each satellite grid box are useful, the features include the grid-box value itself along with the means of 1×1, 9×9, and 15×15 windows of pixels centered on each grid box and are based on three inputs: the 12th channel and the first two components obtained by applying PCA to the other seven IR channels (see Table II).

In addition, the local textural features from cloud patches, which are similar to those used in the PERSIANN-CCS algorithm [16], were also extracted. The ITT algorithm [16] was adopted to determine the cloud patches on the bright temperature data from the 12th channel with three thresholds of 220 K, 235 K, and 253 K. After the patches were determined, two kinds of feature information were extracted: 1) the case where all the pixels' values were the same in each patch with the information of the cloud patch area, minimum and mean values, and standard deviation and 2) the case where each pixel's feature differed according to its relationship to the pixel with the lowest temperature in the corresponding patch, e.g., the distance [see (1)], difference (delta) [see (2)], and slope [see (3)]

$$\text{Distance} = \sqrt{(x_p - x_c)^2 + (y_p - y_c)^2} \tag{1}$$

$$\text{Delta} = T_p - T_c \tag{2}$$

$$\text{Slope} = \text{Delta}/\text{Distance} \tag{3}$$

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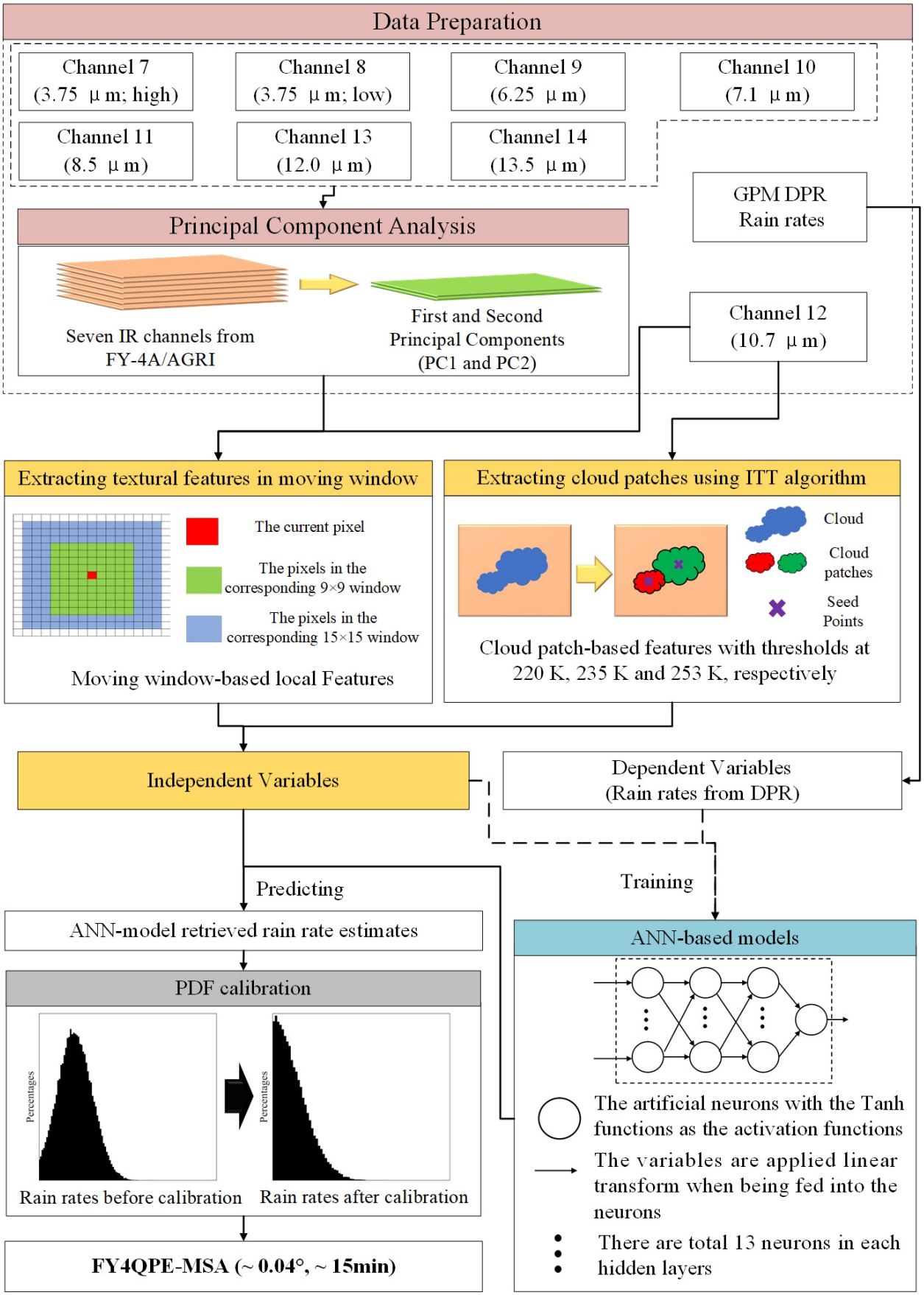

Fig. 1. Flowchart of the framework proposed in this study (FY4QPE-MSA) to retrieve rain rates for Chinese FY-4 series satellites.

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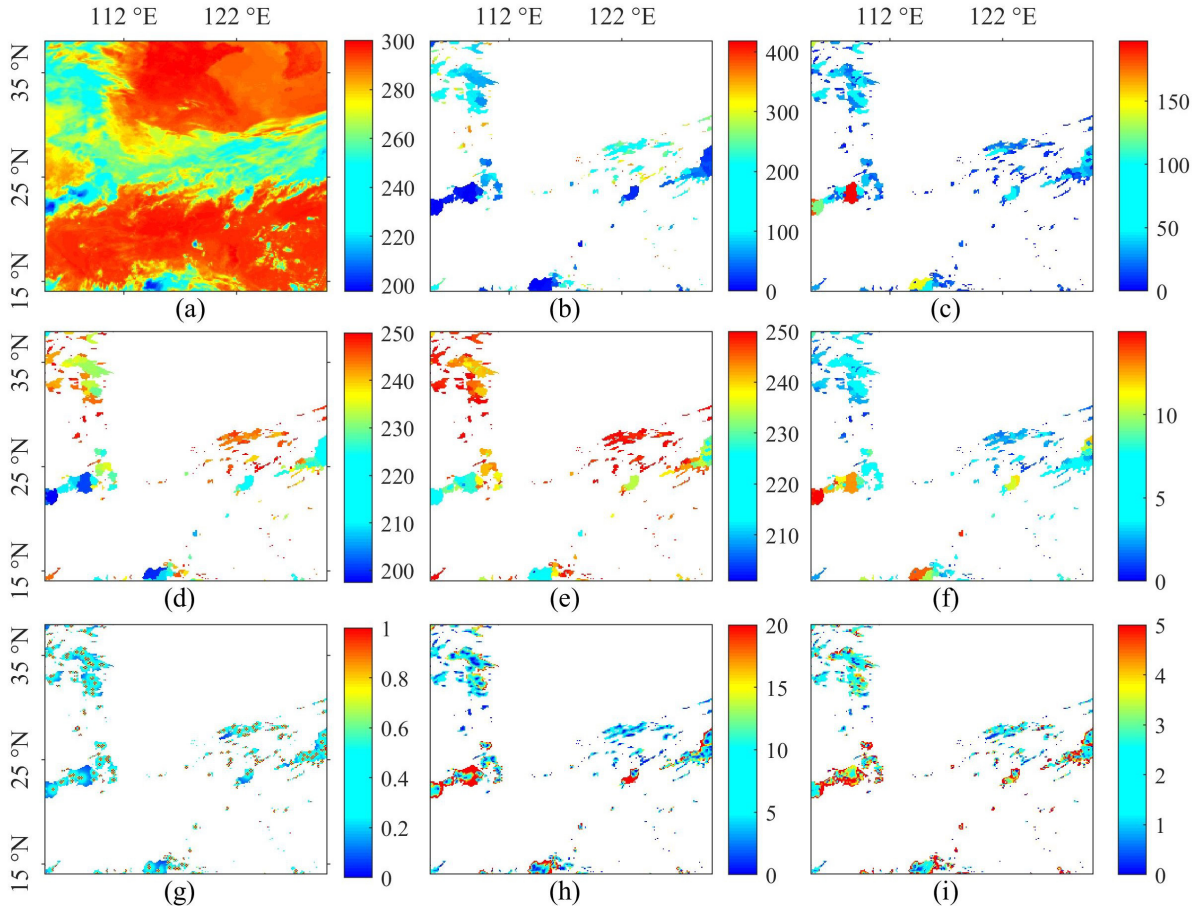

Fig. 2. Spatial patterns of (a) 12th channel from FY-4A/AGRI, (b) patches extracted using ITT algorithm with a temperature threshold of 253 K, and their corresponding patch-based (c) area values, (d) minimum values, (e) mean values, (f) standard deviation values, (g) 1/distance values, (h) difference values, and (i) slope values at a typical hour (00:00–01:00, June 1, 2018).

where x_c, y_c, and T_c represent the longitude, latitude, and bright temperature of the pixel with the lowest temperature value in the coincident cloud patch; similarly, x_p, y_p, and T_p represent the longitude, latitude, and bright temperature of the current pixel in the corresponding patch. All the feature information used in this framework is listed in Table II.

The patches extracted on the 12th channel [see Fig. 2(a)] using the ITT algorithm with a threshold of 253 K are shown in Fig. 2(b) at a typical hour (00:00–01:00, June 1, 2018); these patches are spatially larger than those extracted with thresholds of 220 K and 235 K. The extracted patches are mainly located at the central and northwestern regions, where the brightness temperature values are relatively low in the 12th IR channel; the other corresponding characteristics are shown in Fig. 2(c)–(i). For instance, the patch-based minimum values vary from ∼197 K to ∼253 K (mean value: ∼230 K and standard deviation: ∼15 K), as shown in Fig. 2(d); similarly, the patch-based mean values vary from ∼201 K to ∼253 K (mean value: ∼240 K and standard deviation: ∼ 10K), as shown in Fig. 2(e).

3) Strategy for Finalizing the Training Dataset: The procedure used to finalize the training dataset consisting of the samples with DPR rain rates and the corresponding coincident IR features comprises four main steps. First, all features based on FY-4A/AGRI multichannel IR observations (see Table II) are extracted at the resolutions of 0.04° and 1 h in each season, for example, in summer (June to October) 2019, over the Northern Hemisphere observation range of FY-4A/AGRI (62.0°–135.0° E, 5°–53° N). Second, the observation belts of GPM/DPR onboard the LEO satellite and overlapping with those of FY-4A/AGRI are selected. Third, the samples consisting of point-based DPR rain rates with values of larger than 0.01 mm and the corresponding coincident local features from multi-IR channels (0.04°, 1 h) are spatiotemporally matched, which generates ∼5 million samples in summer 2019, over this study region. Fourth, the samples in each overlapped belt between GPM/DPR and FY-4A/AGRI are ranked according to the DPR rain rates from smallest to largest, and the results are used to determine the samples at the hundred quantiles as the training dataset, reflecting the rain events at various magnitudes. Overall, the final training dataset includes ∼150 000 samples, including ∼140 000 samples with DPR values smaller than 5.0 mm/h and ∼50 000 samples with DPR values larger than 10.0 mm/h. Because one of the objectives of this study is to provide an adaptive framework for training the rain-rate retrieving model at a seasonal scale, the DPR observations also cover the corresponding seasons as

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the FY-4A/AGRI over the Northern Hemisphere observation range.

4) Training the Model: The precipitation retrieval framework is based on the ANN model, which has three hidden layers with 13 neurons in each layer and one output layer (see Fig. 1). In the ANN model, the activation functions of all the neurons in hidden layers are set as tansig, which is forced by 30 features for rain-rate estimation. In addition, the other fundamental parameters in the ANN model are the learning rate, max training epochs, cost goal, and the cost function, which are set as 1, 5000, 0.001, and RMSE, respectively, on the basis of our experience.

Before the ANN-based retrieving model is applied, all of these features and precipitation data should first be normalized. The Z-score strategy is adopted in this study for normalizing these features, which can be described using (4). Because of the highly skewed and tailed characteristics, the normalization strategy for precipitation data is applied using our self-defined approaches shown in (5) and (6)

$$F_1 = (F_0 - \text{mean}(F_0))/\text{std}(F_0) \tag{4}$$

$$P_1 = (P_0 + 1)^{0.5} - 1 \tag{5}$$

$$P_2 = P_1 - \text{mean}(P_1) \tag{6}$$

where F_0 is the original input features and F_1 is the normalized results for input features; similarly, P_0 represents the precipitation data before normalization, and P_2 is the final normalized results, while P_1 represents the intermediate precipitation variable.

5) PDF Calibration: Because of the highly skewed training samples, the original estimates from the ANN model are not suitable for the final retrieved rain rates. Therefore, PDF calibration is necessary to adjust the original estimates from the ANN model for better performances; the calibration is based on strategies that comprehensively consider the differences between the observed rainfall amounts' distributions and the estimates' distributions (e.g., the strategies proposed by Behrangi *et al.* [14] and Tao *et al.* [32]). First, the normalized estimated values (P_2) from the ANN model are transformed into the nonnormalized rain-rate values (P_0) using (5) and (6). Second, among the values estimated from the ANN model, those smaller than 1.5 mm/h are set as zeros to remove the skewness according to the statistical distributions of DPR rain rates from the sample-training ANN model [see (7)]. Finally, the skewness-calibrated estimates are multiplied by the scale coefficient K to stretch them to approach the statistical distributions of the DPR rain rates from the sample-training ANN model. Notably, K is related to the elevations over the land areas by considering the potential negative effects of high elevations [see (8)]. For instance, the cloud temperature over the Tibetan Plateau (average elevation: ~4000 m) is generally smaller than that over the plains, which could easily result in an overestimation of the rain-rate volumes

$$P_{\text{calibrated estimation}} = (P_{\text{model estimation}} - \text{Skewness}) * K(\text{Ele}) \tag{7}$$

$$K(\text{Ele}) = \begin{cases} 5.23, & \text{if Ele} \le 3000 \text{ m} \\ \dfrac{5.23}{e} \cdot e^{-\frac{\text{Ele}-4500}{1500}}, & \text{if Ele} > 3000 \text{ m} \end{cases} \tag{8}$$

where $P_{\text{model estimation}}$ is the original rain rate estimates from the ANN-model and $P_{\text{calibrated estimation}}$ represents the final precipitation estimates after PDF calibration; the Skewness here is set as 1.5 mm/h; meanwhile, K is the function of elevations, and Ele is the abbreviation of the elevation.

C. Evaluation Metrics

To compare the rain-rate retrievals based on observations from FY-4A/AGRI with those based on observations from FY4AQPE-Official and PERSIANN-CCS, five statistical metrics were adopted in this study. To identify the rain occurrences, we used the probability of detection (POD), the false alarm ratio (FAR), and the critical success index (CSI); to evaluate the volumes of the rain rates, the Pearson correlation coefficient (CC) and the root mean square error (RMSE) were applied [30]. To better reveal the differences, the gain/loss index was also applied to compare the metrics of the FY4AQPE-MSA and those of other concurrent IR-based rain-rate estimates, including PERSIANN-CCS, FY4AQPE-Single, and FY4AQPE-Official, according to (9) [14]

$$\%\text{Gain/Loss} = \frac{\text{Metric}_{\text{obp}} - \text{Metric}_{\text{offp}}}{\text{Metric}_{\text{offp}}} \times 100\% \tag{9}$$

where Metric represents the metric (i.e., CC, RMSE, POD, FAR, or CSI), obp denotes the FY4AQPE-MSA, and offp denotes the PERSIANN-CCS, FY4AQPE-Single, or FY4AQPE-Official.

IV. Results

A. Spatial Patterns of FY4AQPE-MSA and Other Products at Different Scales

The spatial patterns of the FY4AQPE-MSA, PERSIANN-CCS, IMERG-LateRun, FY4AQPE-Single, and FY4AQPE-Official over the Northern Hemisphere of FY-4A/AGRI in the period of August 2019 are shown in Fig. 3. In general, the FY4AQPE-Official product [see Fig. 3(e)] could not capture the spatial patterns of the precipitation compared with the two international baseline products: IR-based PERSIANN-CCS [see Fig. 3(b)] and PWM-based IMERG-LateRun [see Fig. 3(c)]. Worse yet, FY4AQPE-Offical appears to be a pixel-based precipitation product that overall underestimates the precipitation volume, especially in the tropical regions (e.g., in the solid and dotted circles). In the solid circles, FY4AQPE-MSA [see Fig. 3(a)] shares two patches with large precipitation intensities, similar to those of the PMW-based IMERG-LateRun, whereas the PERSIANN-CCS and FY4AQPE-Single [see Fig. 3(d)] share one patch with large precipitation volumes. Similarly, in the dotted circles, FY4AQPE-MSA demonstrates much greater similarity with IMERG-LateRun, whereas both FY4AQPE-Single and PERSIANN-CCS underestimate the precipitation volumes. In this situation, the information from MSA is valuable in retrieving IR-based rain rates with fine resolutions.

The spatial patterns of the five IR-based estimates over southeastern China are shown in Fig. 4. At the regional scale, taking the PMW rain estimates as the "observation" [see Fig. 4(c)], we find FY4AQPE-Official [see Fig. 4(e)] still failed

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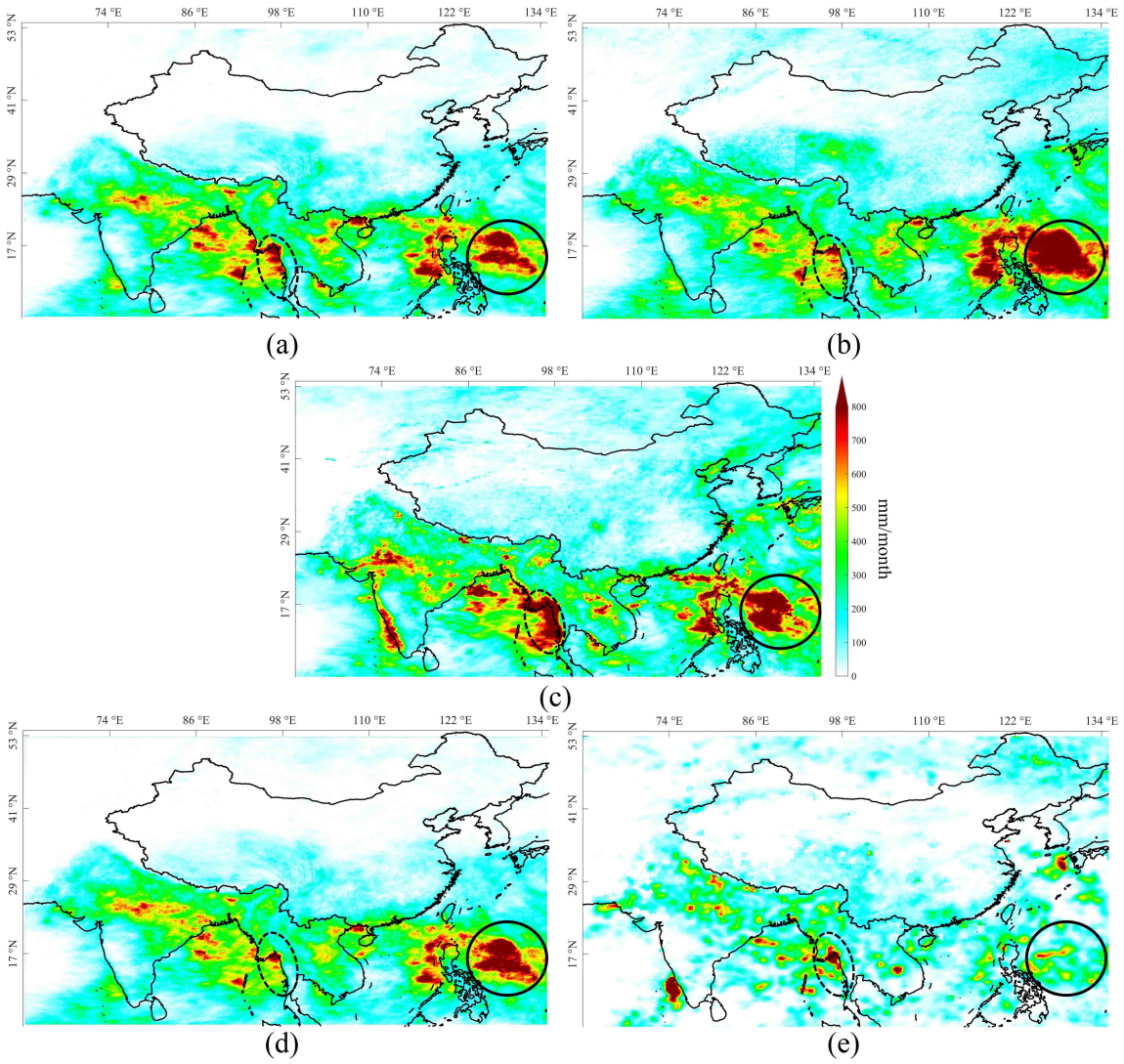

Fig. 3. Spatial patterns of (a) FY4AQPE-MSA, (b) PERSIANN-CCS, (c) IMERG-LateRun, (d) FY4AQPE-Single, and (e) FY4AQPE-Official over the Northern Hemisphere of FY-4A/AGRI in the period of August 2019.

to capture the significant continuous spatial distributions, especially in the southeastern regions with high-intensity precipitation [see Fig. 4(c)]. Although FY4AQPE-MSA [see Fig. 4(a)], FY4AQPE-Single [see Fig. 4(d)], and PERSIANN-CCS [see Fig. 4(b)] demonstrate patterns much more similar to PMW-based estimates [see Fig. 4(c)], FY4AQPE-MSA still has several advantages over PERSIANN-CCS and FY4AQPE-Single. For instance, PERSIANN-CCS overestimates the rain rates in the northwestern regions over the land and in the southeastern areas over the South China Sea (marked by solid rectangles). In addition, with multispectral information, FY4AQPE-MSA outperforms FY4AQPE-Single, providing much more detailed spatial patterns [e.g., in the western regions (marked by the dotted circles)].

A visual event-scale exploration of a precipitation event at a typical hour (13:00–14:00, August 8, 2019) over the Western Pacific Ocean is shown in Fig. 5. FY4AQPE-MSA, PERSIANN-CCS, PMW rain estimates (CLIM-METOPC-MHS-GPROF2017v2), FY4AQPE-Single, and FY4AQPE-Official are shown in Fig. 5(a)–(e), respectively. Overall, by comparison with the PMW-based rain-rate estimates [see Fig. 5(c)], FY4AQPE-MSA outperforms PERSIANN-CCS, FY4AQPE-Single, and FY4AQPE-Official with respect to spatial distribution. For instance, PERSIANN-CCS appears to be relatively overestimating the volumes both in the typhoon's center and the surroundings, and FY4AQPE-Official did not capture the large precipitation intensities over the southern surrounding regions. Especially when comparing the spatial patterns of FY4AQPE-MSA and those of FY4AQPE-Single, the use of additional spectral bands is inferred to lead to important improvements, particularly in the delineation of the areal extent of precipitation.

B. Comparisons of FY4AQPE-MSA and Other Products Against Ground Observations

The spatial distributions of CC and RMSE for FY4AQPE-MSA, FY4AQPE-Single, FY4AQPE-Official, and PERSIANN-CCS against rain-gauge observations,

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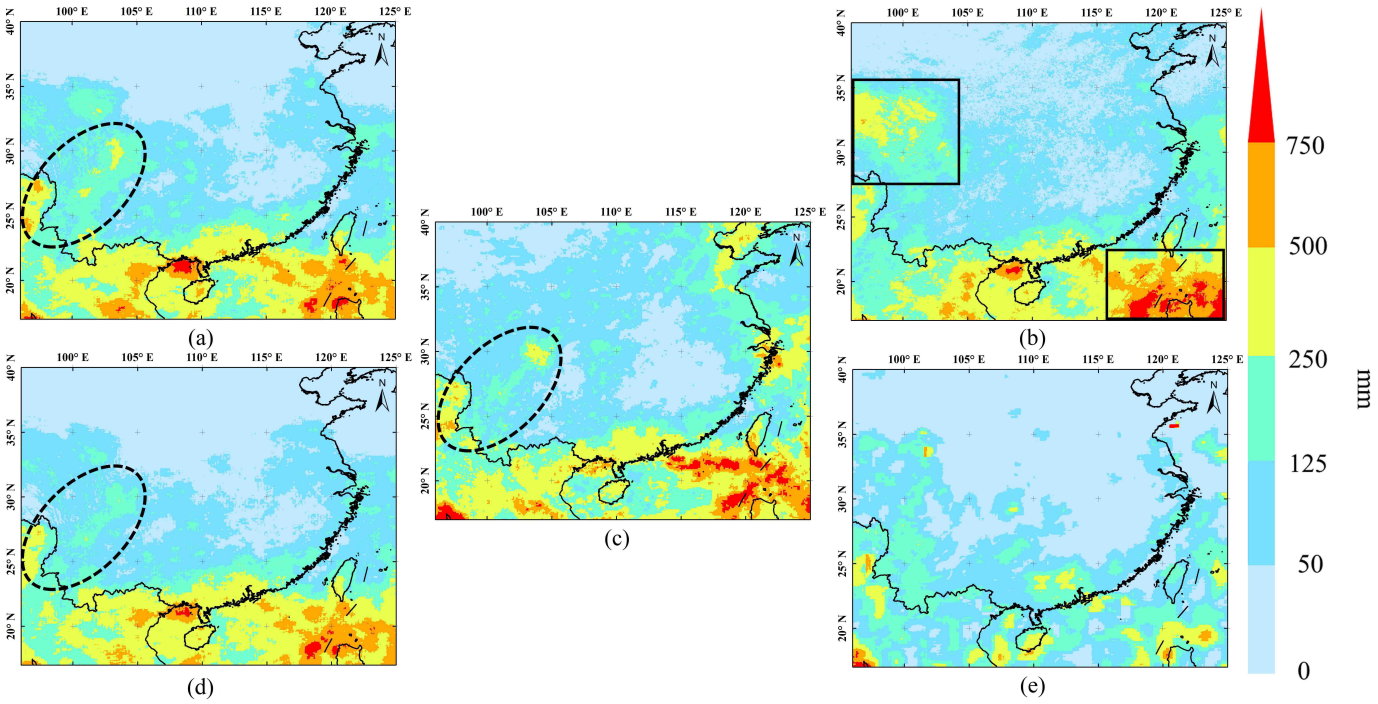

Fig. 4. Spatial patterns of (a) FY4AQPE-MSA, (b) PERSIANN-CCS, (c) IMERG-LateRun, (d) FY4AQPE-Single, and (e) FY4AQPE-Official over southeastern China in the period of August 2019.

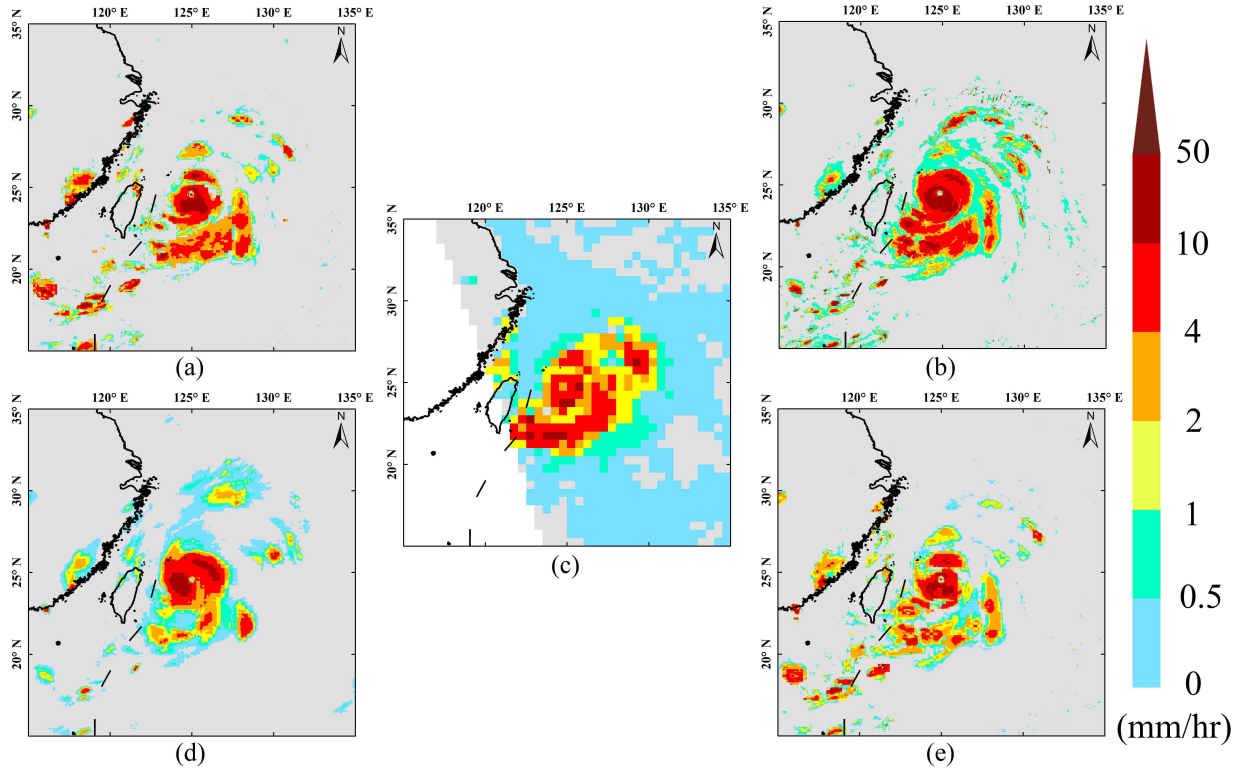

Fig. 5. Comparisons of the rain rate estimation for a precipitation event at a typical hour (13:00–14:00, August 8, 2019) over the Western Pacific Ocean: (a) FY4AQPE-MSA, (b) PERSIANN-CCS, (c) PMW rain estimates (CLIM-METOPC-MHS-GPROF2017v2), (d) FY4AQPE-Single, and (e) FY4AQPE-Official.

in mainland China at the hourly scale in the period from June to September 2019, are shown in Fig. 6. In terms of CC, FY4AQPE-MSA notably outperforms the other three products with the largest areas in the case of CC values larger than 0.3; the performance of FY4AQPE-Single performs approximately the same as that of PERSIANN-CCS, whereas FY4AQPE-Official exhibits the worst performance, especially over southeastern mainland China. Similarly,

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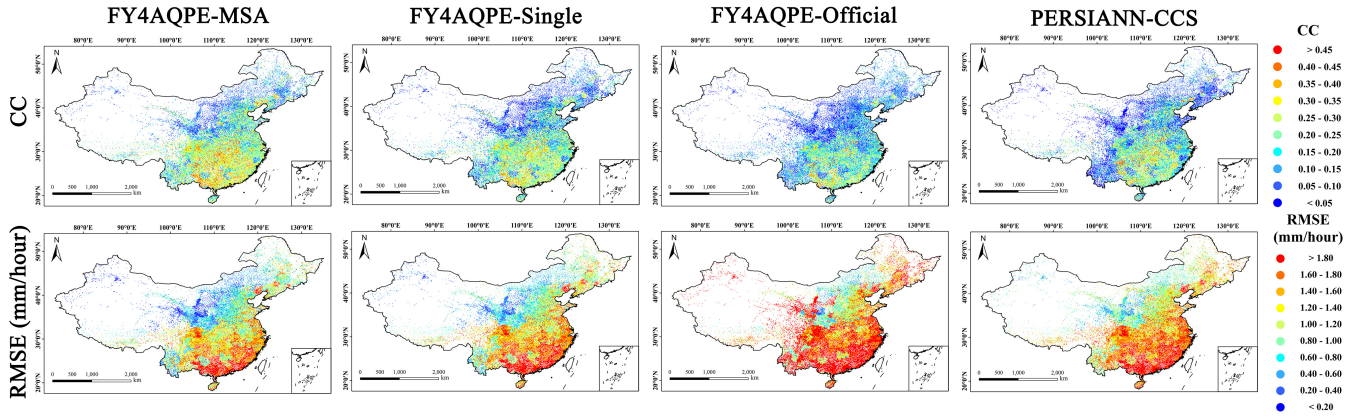

Fig. 6. Spatial distributions of CC (first row) and RMSE (second row) for FY4AQPE-MSA (first column), FY4AQPE-Single (second column), FY4AQPE-Official (third column), and PERSIANN-CCS (fourth column) in mainland China at the hourly scale in the period from June to September 2019.

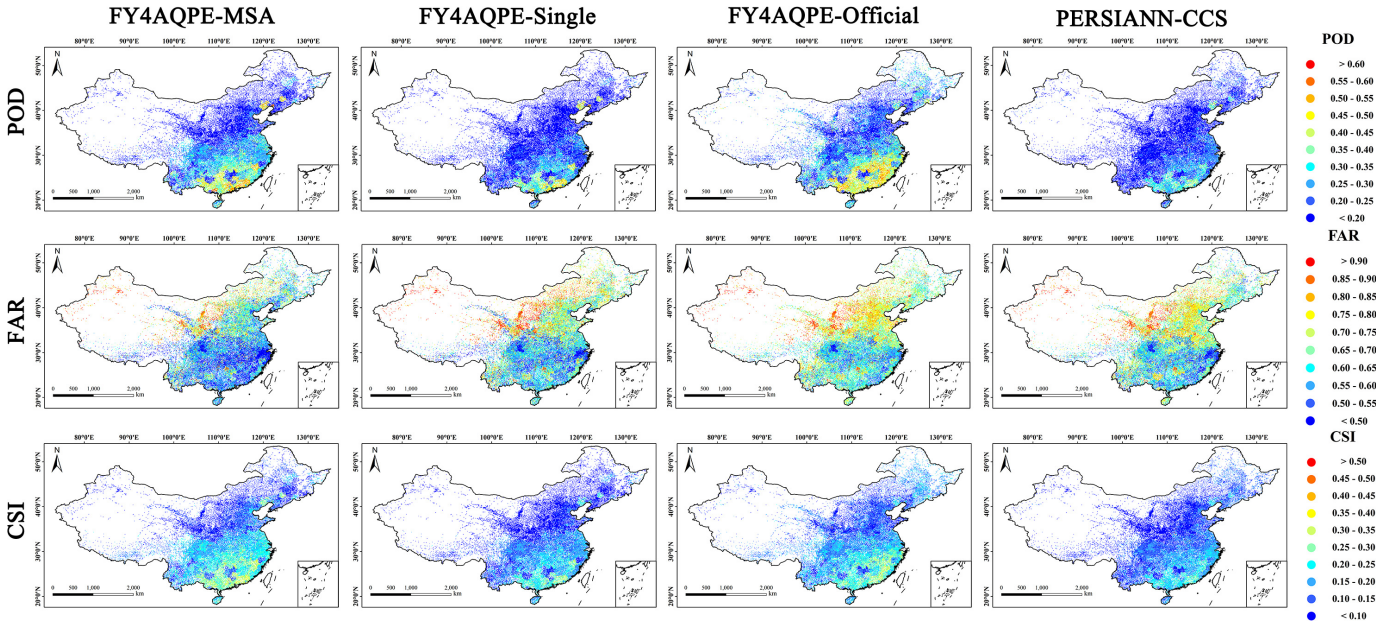

Fig. 7. Spatial distributions of the POD (first row), FAR (second row), and CSI (third row) for FY4AQPE-MSA (first column), FY4AQPE-Single (second column), FY4AQPE-Official (third column), and PERSIANN-CCS (fourth column) in mainland China at the hourly scale in the period from June to September 2019.

the spatial patterns of the RMSE of the four products over the entirety of mainland China also indicate that FY4AQPE-MSA performs best, resulting in the smallest errors, followed by FY4AQPE-Single and PERSIANN-CCS, whereas the FY4QPE-Official demonstrates the relatively largest errors. An interesting phenomenon is that FY4AQPE-MSA outperforms FY4AQPE-Official, with notably smaller errors over western and northern mainland China, where low-intensity small rainfall events tend to occur.

The spatial distributions of the POD, FAR, and CSI for FY4AQPE-MSA, FY4AQPE-Single, FY4AQPE-Official, and PERSIANN-CCS in mainland China at the hourly scale in the period from June to September 2019 are shown in Fig. 7. In terms of POD, FY4AQPE-Official appears to deliver the best performance with most POD values larger than 0.45 over southeastern mainland China; this performance is slightly better than that of FY4AQPE-MSA. FY4AQPE-Official and FY4AQPE-MSA notably outperformed FY4AQPE-Single and PERSIANN-CCS. In terms of the FAR, the performance of FY4AQPE-Official is substantially worse than that of FY4AQPE-MSA, and FY4AQPE-Official appears to be the algorithm most prone to misclassify no-rain events as rain events. Overall, we find that FY4AQPE-MSA performs best in terms of the CSI as a comprehensive index of POD and FAR. In particular, compared with the improvements in the CSI provided by FY4AQPE-Single, those afforded by FY4QPE-MSA provide strong evidence that multispectral information has strong potential for effectively identifying rain-rate events.

To further demonstrate the performance of the FY4AQPE-MSA, the metric "gain/loss" was adopted

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TABLE III

AVERAGE METRICS AND THE PERFORMANCE GAINS OF FY4AQPE-MSA (MSA) AGAINST FY4AQPE-OFFICIAL (OFF), PERSIANN-CCS (CCS), AND FY4AQPE-SINGLE (SIG) IN TERMS OF THE CC, RMSE, POD, FAR, AND CSI AT RAIN GAUGES IN MAINLAND CHINA AT THE HOURLY SCALE FROM 2019 TO 2020 (EXCLUDING DECEMBER, JANUARY, AND FEBRUARY) AND IN SPRING (MAM), SUMMER (JJA), AND AUTUMN (SON) OF 2019 AND 2020, RESPECTIVELY. NOTABLY, NEGATIVE VALUES FOR THE PERFORMANCE GAINS OF RMSE AND FAR REPRESENT POSITIVE IMPROVEMENTS IN PERFORMANCES. THE BEST VALUES ARE SHOWN IN BOLD

Products	Metrics	CC /Gain	RMSE (mm/h)/Gain	POD /Gain	FAR /Gain	CSI /Gain
2019–2020	MSA	**0.20**	**0.92**	**0.21**	**0.64**	**0.15**
	SIG	0.18/11.1%	1.09/-15.6%	0.18/16.7%	0.67/-4.5%	0.13/15.4%
	OFF	0.17/17.6%	1.23/-25.2%	0.20/5.0%	0.70/-8.6%	0.13/15.4%
	CCS	0.15/33.3%	1.20/-23.3%	0.16/31.3%	0.71/-9.9%	0.12/25.0%
2019 MAM	MSA	**0.16**	**0.66**	**0.17**	**0.68**	**0.12**
	SIG	0.15/6.7%	0.68/-2.9%	0.15/13.3%	0.72/-5.6%	0.11/9.1%
	OFF	0.15/6.7%	0.72/-8.3%	0.14/21.4%	0.74/-8.1%	0.10/20.0%
	CCS	0.10/60.0%	0.97/-32.0%	0.14/21.4%	0.77/-11.7%	0.10/20.0%
2019 JJA	MSA	**0.22**	**1.08**	0.27	**0.59**	**0.19**
	SIG	0.20/10.0%	1.24/-12.9%	0.22/22.7%	0.63/-6.3%	0.16/18.8%
	OFF	0.19/15.8%	1.64/-34.1%	**0.29**/-6.9%	0.67/-11.9%	0.18/5.6%
	CCS	0.16/37.5%	1.62/-33.3%	0.20/35.0%	0.67/-11.9%	0.14/35.7%
2019 SON	MSA	**0.17**	**0.49**	0.16	**0.66**	**0.12**
	SIG	0.15/13.3%	0.55/-10.9%	0.15/6.7%	0.71/-7.0%	0.11/9.1%
	OFF	0.13/30.8%	0.57/-14.0%	**0.18**/-11.1%	0.74/-10.8%	**0.12**/0.0%
	CCS	0.10/70.0%	0.71/-40.0	0.11/45.5%	0.74/-10.8%	0.08/50.0%
2020 MAM	MSA	**0.16**	**0.65**	**0.20**	**0.70**	**0.14**
	SIG	0.14/14.3%	0.66/-1.5%	0.18/11.1%	0.73/-4.1%	0.12/16.7%
	OFF	0.14/14.3%	1.05/-38.1%	0.19/5.3%	0.75/-6.7%	0.12/16.7%
	CCS	0.10/60.0%	0.96/-32.3%	0.16/25.0%	0.76/-7.9%	0.11/27.3%
2020 JJA	MSA	**0.25**	**1.32**	0.26	**0.58**	**0.19**
	SIG	0.23/8.7%	1.43/-7.7%	0.23/13.0%	0.61/-4.9%	0.17/11.8%
	OFF	0.21/19.0%	1.81/-27.1%	**0.27**/-3.7%	0.64/-9.4%	0.18/5.6%
	CCS	0.16/56.3%	1.71/-22.8%	0.22/18.2%	0.66/-12.1%	0.15/26.7%
2020 SON	MSA	**0.16**	**0.60**	**0.15**	**0.68**	**0.11**
	SIG	0.14/14.3%	0.62/-3.2%	0.13/15.4%	0.70/-2.9%	0.10/10.0%
	OFF	0.13/23.1%	0.79/-24.1%	0.13/15.4%	0.72/-5.6%	0.10/10.0%
	CCS	0.10/60.0%	0.81/-25.9%	0.11/36.4%	0.72/-5.6%	0.09/22.2%

to calculate the improvement of the estimates of FY4AQPE-MSA at rain gauges compared with the concurrent IR-based estimates. For instance, the CC values were first calculated at the corresponding rain gauges of each precipitation estimate, and the gain/loss values, in terms of CC, were then generated for FY4AQPE-MSA against FY4AQPE-Single, FY4A-Official, and PERSIANN-CCS. Table III summarizes the average metrics of FY4AQPE-MSA, FY4AQPE-Single, FY4AQPE-Official, and PERSIANN-CCS over mainland China from 2019 to 2020 and in spring [March, April, May (MAM)], summer [June, July, August (JJA)], and autumn [September, October, November (SON)] of 2019 and 2020, respectively, as well as the performance gains of FY4AQPE-MSA against FY4AQPE-Single, FY4AQPE-Official, and PERSIANN-CCS in the corresponding temporal periods. Generally, all the products perform notably better in summer than those in spring and autumn. Overall, FY4AQPE-MSA almost performs the best among the four products in all seasons, followed by FY4AQPE-Single, FY4AQPE-Official, and PERSIANN-CCS. In terms of statistical metrics, FY4AQPE-MSA demonstrates the largest gains of ∼70.0% in CC and ∼40% in RMSE against PERSIANN-CCS, in autumn 2019. It should be noted that FY4AQPE-MSA also shows the best performances in accurately identifying the rain occurrences. For instance, FY4AQPE-MSA achieves the largest CSI values in two summers from 2019 to 2020 and the largest performance gain ∼50.0% against PERSIANN-CCS in autumn 2019.

V. DISCUSSION

A. Potential of the Multichannel Observations From the FY-4A/AGRI for Rain Estimation

As previously noted, the traditional and classical approaches to retrieving rain rates mainly focus on the IR band centered at ∼10.7 μm [16], [28], [34]. Under the development of the satellite techniques, the Chinese meteorological satellite sensor FY-4A/AGRI provides eight IR channels for monitoring different physical parameters; maximizing the advantage of these observations is critical to retrieving rain rates with higher quality and greater spatiotemporal resolutions. The spatial patterns of the eight IR observations from FY-4A/AGRI at a typical hour (00:00–01:00, June 1, 2018) are shown in Fig. 8. Though some of the channels demonstrate similar spatial patterns [e.g., channels 7 and 8 (mean value: ∼293 K and standard deviation: ∼10.6 K), channels 10 and 14 (mean value: ∼252 K and standard deviation: ∼8 K), and channels 11–13 (mean value: ∼280 K and standard deviation: ∼17.0 K)], they provide much more information than the IR channel centered only at ∼10.7 μm, both for spatial distributions and volumes [e.g., channel 9 (mean value: ∼238 K and standard deviation: ∼5.5 K)]. Therefore, PCA is applied to extract the greatest

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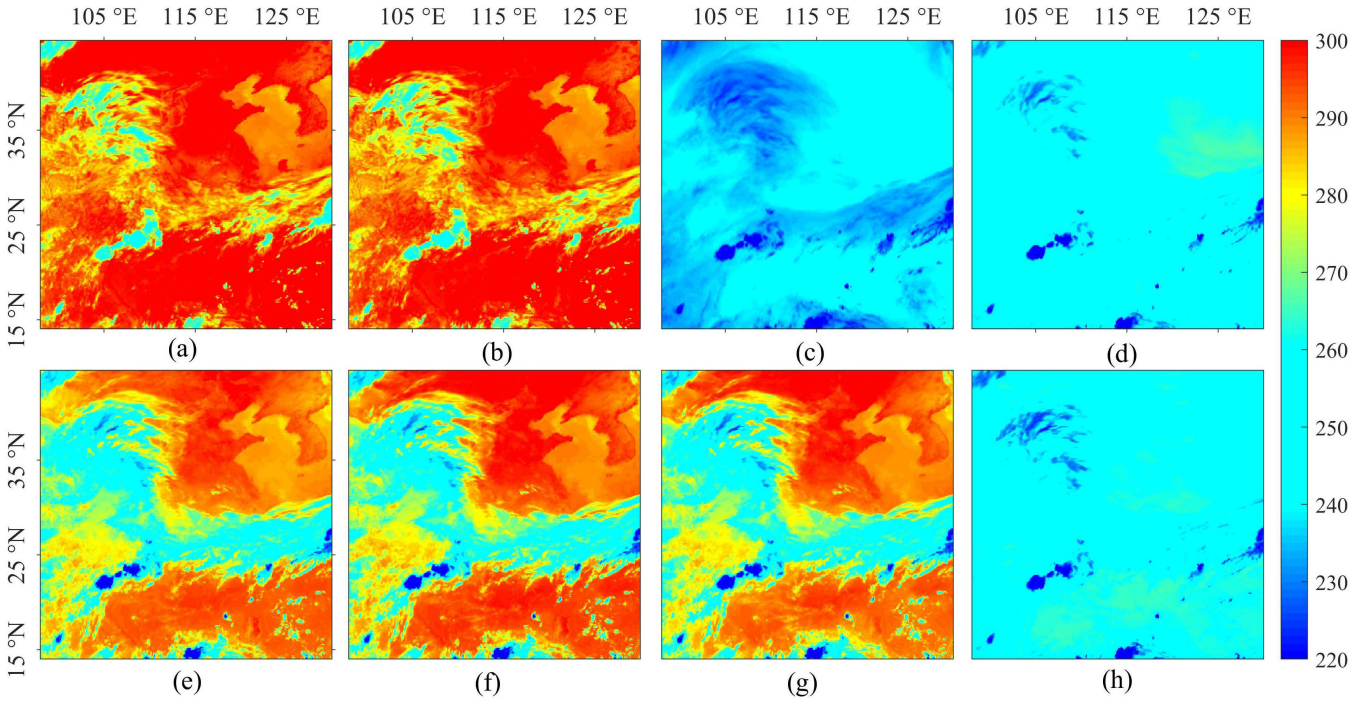

Fig. 8. Spatial patterns of the eight IR observations from FY-4A/AGRI at a typical hour (00:00–01:00, June 1, 2018). (a) Channel 7. (b) Channel 8. (c) Channel 9. (d) Channel 10. (e) Channel 11. (f) Channel 12. (g) Channel 13. (h) Channel 14.

TABLE IV

COEFFICIENTS TO OBTAIN THE PRINCIPAL COMPONENTS BASED ON THE OTHER SEVEN IR CHANNELS FROM JUNE TO SEPTEMBER 2018 USING THE PCA TECHNIQUE

	PCA1	PCA 2	PCA 3	PCA 4	PCA 5	PCA 6	PCA 7
Channel 7	0.371	-0.482	0.356	0.059	-0.001	-0.037	0.706
Channel 8	0.371	-0.481	0.358	0.051	-0.017	0.025	-0.707
Channel 9	0.357	0.580	0.441	-0.085	0.571	0.091	-0.001
Channel 10	0.377	0.431	0.171	-0.026	-0.797	-0.086	0.006
Channel 11	0.390	-0.078	-0.429	-0.456	-0.008	0.670	0.032
Channel 13	0.391	-0.035	-0.408	-0.349	0.168	-0.727	-0.032
Channel 14	0.388	0.081	-0.414	0.810	0.104	0.072	-0.003
Contributions	90.59%	6.71%	2.00%	0.42%	0.18%	0.09%	0.00%

variation with a minimum number of independent components based on the seven IR channels other than the classical band at $\sim$10.7 μm.

Taking the period from June to September 2018 as an example, we used all of the IR channels, except the 12th channel to generate the PCA coefficients to determine the major components representing the greatest variations. As shown in Table IV, the first two components, including the most variations of the other seven IR channels ($>$97%) and their coefficients, were, finally, applied to extract the first two components on the basis of the multichannel IR observations from FY-4A/AGRI in the prediction period (August 2019). The effectiveness of the PCA technique in compressing the number of original input features into fewer features while preserving most of the total variations is helpful for reducing the computational burden and accelerating the rain-rate estimating process, which is particularly important when the framework is designed for operational purposes [14].

The spatial patterns of the 12th channel, along with the first two principal components at the corresponding typical hour (00:00–01:00, June 1, 2018), are shown in Fig. 9. The spatial patterns of the 12th channel are similar to those of the first principal component, which further confirms the rationality of using the band centered at $\sim$10.7 μm for retrieving the rain rates. However, the spatial patterns of the 12th channel differ substantially from those of the second principal component, illustrating the excellent potential for using more information from other IR channels.

B. Potential Advantages of FY-4 Series Satellites Over the Asia–Pacific Region

The original observations of the international IR-based PERSIANN-CCS come from three satellites: the Geostationary Operational Environmental Satellite (GOES-R, NOAA), Himawari-8/9 (JMA), and Meteosat Third Generation (EUMETSAT) satellites [35]. We find an obvious spatial gap in the transition from west to east along the longitude at $\sim$90° E over the Tibetan Plateau in PERSIANN-CCS [see Fig. 10(b)], which is a result of the spectral differences caused

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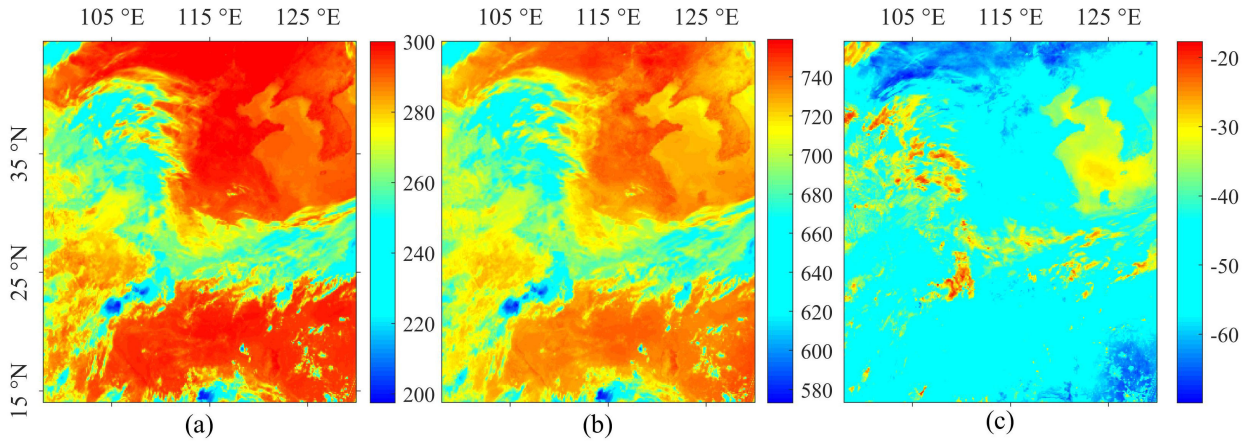

Fig. 9. Spatial patterns of (a) 12th channel, (b) first principal component, and (c) second principal component at a typical hour (00:00–01:00, June 1, 2018).

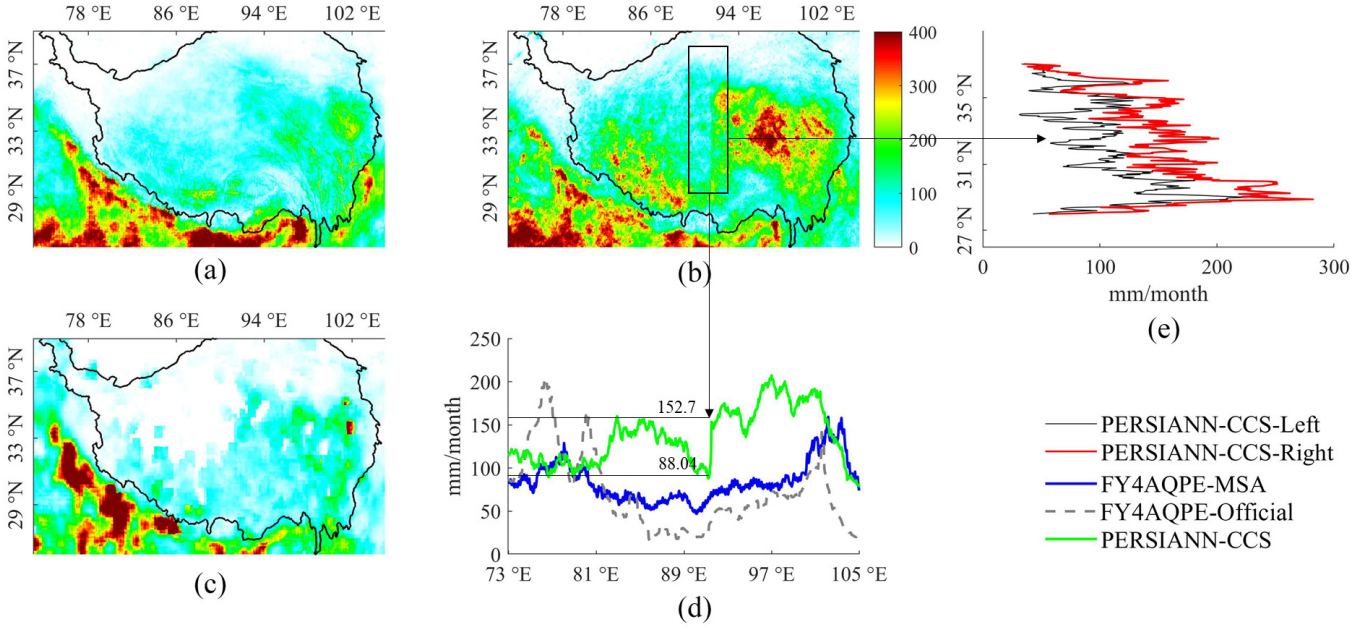

Fig. 10. Spatial gap of precipitation estimation in PERSIANN-CCS at ∼90° E, both in latitudinal and longitudinal directions, in August 2019. (a) FY4AQPE-MSA. (b) PERSIANN-CCS. (c) FY4A-Official. (d) Average rain rates along the longitude. (e) Rain rate differences along with the mosaic.

by different sensors from different satellites (e.g., the Flexible Combined Imager (FCI) and AHI onboard the European Meteosat Third Generation satellites and Himawari-8/9 satellites, respectively). In the longitudinal direction, a ∼70-mm increase in rain volumes is observed in August 2019 [see Fig. 10(d)], whereas, in the latitudinal direction, the estimates in the eastern parts are approximately twice as large as those in the western parts along the longitude at ∼90° E. However, such spatial gaps do not exist in the precipitation estimates of FY4AQPE-MSA [see Fig. 10(a)] and FY4A-Official [see Fig. 10(c)] over the Tibetan Plateau. Therefore, the advantages of the Chinese GEO FY-4 series satellites should be further developed, especially for the Asia–Pacific regions.

C. Possible Future Directions for Retrieving More Accurate All-Day Near-Real-Time Rain Rates

Multichannel IR observations carry different information about cloud properties, which could together facilitate more accurate retrieval of the rain rate, especially for delineating the spatial extents of rain events, compared with those based on the single classical IR band centered at ∼10.7 μm [14]. However, the obtained cloud properties based on these multichannel IR observations are mainly limited to the cloud-top information, which is one of the critical factors restricting the accuracy of IR-based rain-rate retrievals [16]. Historically, passive and active microwave sensors have provided the greatest contribution to the retrieval of much more accurate satellite-based precipitation estimates, which are now only available from LEO platforms, and these sensors will continue in this role in the foreseeable future. This prediction is based on the advantages of using microwave observations to measure microphysical information, including both liquid and frozen hydrometeors within the clouds in the atmospheric column by sensing the emissions and scattering of precipitation-sized particles [12].

This study explores a general framework for retrieving all-day near-real-time rain rates using multichannel IR

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observations from FY-4A/AGRI at the resolutions of 0.04° and 1 h. There is much room for improving the capability of this framework to generate precipitation estimates at finer resolutions (∼2 km, ∼5 min), especially in the era of the FY-4 subsequent satellites (e.g., the first operational GEO satellite of FY-4 and FY-4B [June 3, 2021, located at 123.5° E)]. In addition, the possible directions for progressively enhancing this framework should include, but not be limited to, the following aspects: 1) considering more physical-based processes (e.g., the mechanisms that form the convective and strigiform precipitation events); 2) further mining the values of other cloud microphysical properties (such as CTT, CTH, CP, and WV) [37]; 3) comprehensively utilizing the advantages of PMW observations from both international satellites and Chinese meteorological satellites [e.g., Chinese Fengyun-3 series LEO satellites, especially the FY-3E (launched on July 5, 2021)]; and 4) introducing other more comprehensive and robust deep learning methods, such as CLSTM, CNN, GAN, and DBN mentioned in Section I.

VI. Conclusion

Accurate and near-real-time rain information at fine scales is critical for forecasting local weather and floods to protect human life and property. Given that the IR cloud-top brightness temperature data alone do not contain sufficient precipitation-related information, one possible direction is to explore the benefits of multispectral channels from GEO satellites for rain detection and estimation. However, the common choices using VIS cloud albedo data are usually limited to daytime. Therefore, explorations of how to efficiently and comprehensively utilize multichannel IR observations are an inevitable trend for accurately retrieving near-real-time rain rates with consistent high quality for all-day operational purposes. This ability would be particularly useful in the era of GEO satellites capable of enhanced spatial–temporal–spectral resolution measurements, e.g., the FY-4 series subsequent satellites and sensors.

In this study, we proposed an all-day near-real-time QPE framework based on MSA from AGRI onboard Chinese FY-4 series satellites; we refer to this framework as FY4QPE-MSA. We comprehensively and efficiently considered the multispectral IR bands by adopting the PCA technique to reduce the dimensionality to a few independent input features while preserving most of the variations. The main conclusions include, but are not limited to, the following aspects: 1) the MSA from IR observations provides valuable information that enables more accurate delineations of the precipitation occurrences; 2) FY4AQPE-MSA outperforms FY4AQPE-Offical (∼20% gain in CC, ∼25% gain in RMSE, and ∼15% gain in CSI), FY4AQPE-Single (∼10% gain in CC, ∼15% gain in RMSE, and ∼15% gain in CSI), and PERSIANN-CCS (∼30% gain in CC, ∼20% gain in RMSE, and ∼25% gain in CSI); and 3) compared with the international IR-based baseline precipitation product, PERSIANN-CCS, FY4AQPE-MSA demonstrates no spatial gaps over the Tibetan Plateau. Therefore, the advantages of the Chinese GEO FY-4 series satellites should be further developed, especially for the Asia–Pacific regions. In addition, the results of this study suggest that the framework is promising and applicable for Chinese FY-4 series subsequent satellites in generating rain estimates with finer quality and resolutions.

Competing Interests

Competing Financial Interests: The authors declare no competing financial interests.

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